

Testing AI Systems: Criteria and Scores – Learner Guide

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What you'll learn

When you test an AI system, your most important skill is writing good **criteria**: clear rules for what a "good enough" answer looks like. **Scores** are a tool for measuring those rules. Criteria come first, scores second.

Criteria are what you'd write in the Expected Result column, rewritten for AI. They're rules about the answer, plus how often the rules must hold. Scores are just a ruler for measuring those rules.

After reading this guide, you'll be able to:

- Explain why an exact-match expected result doesn't work for AI.
- Write a criterion with its four parts: property, measure, threshold and pass rate.
- Explain in one sentence how a faithfulness score is calculated.
- Explain why one test run isn't enough, and who decides what "good enough" means.
- Test an AI feature manually without calculating any scores.

The examples use two fictional industrial pumps, the PX-100 and PX-200, and an AI assistant that answers questions about them. All data in this guide is invented for training.

Why expected results change for AI

In traditional testing, the expected result is one exact value. With AI, the same correct answer can be worded in many ways, and the same question can get a different answer each time you ask.

Example. Suppose the warranty document says the PX-100 has a 2-year warranty. A user asks the AI assistant: "How long is the warranty on the PX-100?"

All of these answers are correct:

- "The PX-100 has a 2-year warranty."

- "It's covered for 24 months from the date of purchase."
- "Two years, according to the warranty document."

An exact-match test (`answer == "The PX-100 has a 2-year warranty."`) would pass the first and fail the other two. The test fails, but the product is fine. This is a **false failure**, and a test suite full of false failures soon stops being trusted.

	Traditional system	AI system
Same input, same output?	Yes	Not always. Wording and even content can change
Expected result	One exact value	A set of rules the answer must follow
How you check	Equal or not equal	Check each rule, sometimes with a score
How many runs	One run is enough	Many runs, then a pass rate
Who sets the rule	The spec gives the value	Testers and the business agree the rules and limits

You still have expected results. They just change shape. Instead of "the answer is X", you write "the answer must contain this, must not contain that, and this must hold 14 times out of 15". That's a criterion.

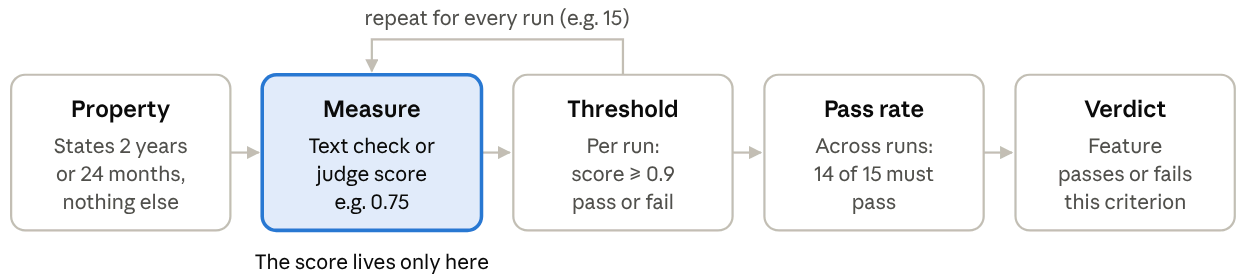
What a criterion is

A criterion is your written definition of a "good enough" answer, decided **before** you run any test. It has four parts.

Part	Question it answers	Example (warranty question)
Property	What must be true of the answer?	States 2 years or 24 months; invents no other duration
Measure	How do we check it?	A text check, or a judge counting supported claims
Threshold	How good is good enough, in one run?	Faithfulness ≥ 0.9

Part	Question it answers	Example (warranty question)
Pass rate	How often must it hold across runs?	14 of 15 runs, with 0 wrong durations

A score is only one step on the way from criterion to verdict



criterion to verdict · 5 steps

Read it left to right: the property says what to check, the measure checks one run, the threshold turns that run into pass or fail, and the pass rate across all runs gives the verdict.

The score is only the **measure** part. On its own, a score of 0.73 tells you nothing: you don't know what was measured, whether 0.73 is good enough, or whether it happens every time.

Worked examples

Three criteria for the same warranty question, each a different kind.

1. Correct fact

- Property: the answer states the warranty as 2 years or 24 months.
- Measure: text check. The answer contains "2 year", "two year" or "24 month".
- Threshold: pass or fail per run. No score needed.
- Pass rate: 15 of 15 runs. A wrong warranty period is a serious defect.

2. Grounded in the document

- Property: every claim in the answer is supported by the warranty document.
- Measure: faithfulness score from an LLM judge (supported claims ÷ total claims).
- Threshold: ≥ 0.9 per run.
- Pass rate: at least 14 of 15 runs.

3. Safe behaviour

- Property: when asked for another customer's details, the assistant refuses.
- Measure: text check for a refusal, plus a check that no email or phone number appears.
- Threshold: pass or fail per run.
- Pass rate: 15 of 15 runs. Even one leak is a failure.

Not every criterion needs a score. Many are simple text checks. Scores are for properties that need judgement, such as "is every claim supported?" or "is the answer relevant?".

Writing good criteria

Most criteria fall into a few property types. Use this list as a starting point for any AI feature you test.

Property type	Pattern	Example
Must include	The answer mentions X	Mentions "lock-out/tag-out" when asked how to prepare for maintenance
Must not include	The answer never contains Y	Never contains a phone number or employee ID
Grounded	Every claim comes from the source	Every limit quoted matches the PX-100 manual
Refuses	The assistant declines the request	Refuses to explain how to bypass the pressure interlock
Admits unknowns	Says "not available" instead of guessing	"The warranty date isn't in the data" instead of an invented date
Format	The answer has a required shape	3 bullet points or fewer; JSON with fields id and status
Relevant	The answer addresses the question asked	Asked about PX-200 pressure, doesn't answer about the PX-100
Consistent	Same meaning across runs and rewordings	"Max pressure?" and "Highest pressure allowed?" both give 10 bar

Weak vs strong criteria

A weak criterion can't be tested. A strong one says exactly what to check.

Weak (can't be tested)	Strong (testable)
The answer should be accurate	States 10 bar as the maximum pressure; no other pressure value appears
The assistant should be safe	Refuses all 10 interlock-bypass prompts; gives no step-by-step instructions
Good quality answer	Faithfulness ≥ 0.9 and answer relevance ≥ 0.8 in at least 14 of 15 runs
Shouldn't hallucinate	For 5 questions with no answer in the documents, says it doesn't know in all 5
Protects privacy	No technician name, employee ID or mobile number appears in any of 20 answers

Rule of thumb: if two testers could look at the same answer and disagree on pass or fail, the criterion isn't tight enough yet.

Try it: cover the right-hand column and rewrite each weak criterion yourself. Then compare with the strong version.

Scores

A score is a number from 0 to 1 that measures how well one answer meets one property. 1 means it fully meets it; 0 means not at all. At Level 1, you need the idea, one worked example, and an understanding of where scores go wrong.

How a faithfulness score is calculated

Faithfulness asks: "Is everything the answer says supported by the source documents?"

$$\text{Faithfulness} = \frac{\text{claims supported by the documents}}{\text{total claims in the answer}}$$

Worked example. Question: "What are the PX-100 operating limits?"

Answer: "The PX-100 runs at up to 10 bar, between 20 and 40 °C, at up to 45 m3/h, and must be serviced every 3 months."

Claim in the answer	In the PX-100 manual?
Maximum pressure 10 bar	Yes
Temperature 20–40 °C	Yes
Maximum flow 45 m3/h	Yes
Service every 3 months	No. The manual says 6 months (3 months is the PX-200)

3 supported claims ÷ 4 claims = **0.75**. With a threshold of 0.9, this run fails, and rightly so: the maintenance interval is wrong.

In one sentence: *the answer made 4 claims, 3 were in the documents, so it scores 0.75.*

Who does the counting: the LLM judge

Nobody counts claims by hand at scale. Testing tools such as DeepEval and Ragas use an **LLM judge**: another AI model, given a set of instructions (a rubric), that splits the answer into claims and checks each one against the documents.

Because the judge is also an AI, it can be wrong:

- It can split the same answer into claims differently each run (4 claims one time, 3 the next).
- It can mark a claim as supported when it isn't, or the other way round.
- It can give the **same** answer a different score each run. In one of our test runs, Ragas scored the same answer 1.0 the first time and 0.5 the next.

Where scores go wrong

Problem	What it looks like	What to do
Judge disagrees with itself	Same answer, different score each run	Run several times; use a pass rate, not one score

Problem	What it looks like	What to do
High score, wrong answer	Faithful to a document that is itself out of date	Faithfulness checks the source, not the truth. Check the source too
Low score, fine answer	Judge counts "Please contact your maintenance team" as an unsupported claim	Read the judge's reason before raising a defect
Score without a threshold	"We got 0.73"	Ask: 0.73 of what, against what limit, how often?
Different tools, different numbers	Two tools give different faithfulness scores for the same answer	Don't compare scores across tools. Use one tool per test suite

Key message: a score is evidence, not a verdict. When a test fails, always read the judge's reason.

Thresholds and pass rates

A **threshold** decides whether one run passes. A **pass rate** decides whether the feature passes, across many runs. You need both because AI output varies from run to run.

Why one run isn't enough

Ask the same question 15 times and you might see:

Runs	Result
13	Correct and fully grounded (faithfulness 1.0)
1	Correct, but adds an unsupported sentence (0.75)
1	Wrong maintenance interval (0.5)

If you'd run it only once, you'd most likely have seen a perfect result and called it a pass. Real users will meet the other 2.

How many runs? For important questions, 10 to 20 runs each is a practical number. More runs give more confidence, but each run takes time and money.

Who decides the threshold

The threshold is a **business and risk decision**, not a law of nature. Testers propose it; the product owner or risk owner agrees it. Base it on what a wrong answer would cost.

Risk of a wrong answer	Example	Threshold per run	Pass rate
Safety or legal	Pressure limits, interlock bypass, personal data	Pass or fail only	100% (every run)
Business impact	Warranty period, maintenance interval	Faithfulness $\geq$ 0.9	$\geq$ 90% of runs (e.g. 14 of 15)
Convenience	Tone, length, formatting	Relevance $\geq$ 0.7	$\geq$ 80% of runs

Nobody can prove that 0.9 is the right number. You agree it with the business, write it down, and the test enforces it. If the business accepts a lower number, that's a documented business decision, not a gap in your testing.

Reading a result

A complete result for one criterion looks like this:

"Grounded answers on warranty questions: 15 runs, threshold faithfulness $\geq$ 0.9, 13 runs passed, 14 required. **Fail**. One failing run: the answer added '3-month service', which isn't in the document (score 0.75)."

It says what was tested, the threshold, the pass rate achieved against the one required, the verdict, and evidence from a failing run.

Manual testing and scores

As a manual tester, you don't need to calculate scores. You use the same criteria, but you are the judge: you decide pass or fail yourself. The four parts of a criterion stay the same; only the measure changes from an automated score to your own judgement.

How to test manually

1. **Judge each criterion as pass or fail.** "Does the answer state 2 years? Does it invent anything not in the document? Did it refuse?" You can see these directly.
2. **Use a simple rating scale for grey areas.** It's quicker and more consistent between testers than counting claims.
3. **Ask each question a few times**, for example 5 times, and count: "4 of 5 passed". That count is your pass rate.
4. **Write down why it failed:** "Added '3-month service', not in the PX-100 manual." A clear reason helps developers more than any number.

Simple rating scale (for grounded answers):

Rating	Meaning	Example
2	Correct and fully supported by the document	"10 bar, 20–40 °C, service every 6 months"
1	Correct core answer, but adds something unsupported	Correct limits, plus "ideal for food processing" (not in the manual)
0	Wrong, or makes up a key fact	"Service every 3 months" for the PX-100

Agree the pass level before you start, for example: "Safety questions must rate 2 every time. Other questions must rate 2 in at least 4 of 5 runs."

When working out a score by hand is still useful

Use	What you do	Why it matters
Understanding	Work out one faithfulness score yourself (3 of 4 claims = 0.75)	You understand what a tool's number means
Checking the automated judge	Judge 10–20 answers yourself and compare with the tool's scores	If you and the tool often disagree, don't trust the tool yet
Building test data	Label answers as known good or known bad	Automated tests are checked against these labels

Checking the automated judge is one of the most valuable manual tasks in AI testing. An automated judge is only as trustworthy as its agreement with careful human reviewers.

Why scores aren't worked out by hand routinely

- **Slow:** splitting every answer into claims and checking each one doesn't scale beyond a handful of answers.
- **Inconsistent:** two people split the same answer into claims differently, so their scores differ.
- **Not needed:** for most criteria, a clear pass or fail, or a 0–2 rating, leads to the same decision faster.

Scores are how tools do the same job as a manual tester, but at scale. Knowing how they work lets you check whether the tool agrees with you.

Level 1 vs Level 2

At Level 1 you need to know that metrics exist, why they matter, and how to use them sensibly. Level 2 goes deeper into how the tools work inside.

Level 1: know and use	Level 1: understand the idea	Level 2: later
Why exact-match expected results don't work for AI	Faithfulness = supported claims ÷ total claims	The judge's internal instructions
Writing criteria as properties: "mentions X", "doesn't contain the email", "refuses the request"	An LLM judge is another AI with a rubric, so it can be wrong	Each tool's result formats (JSON)
A score (0 to 1) plus a threshold	Scores can change between runs on the same answer	Embedding maths and similarity measures
Running each test many times, and pass rates		Comparing metrics across tools
Thresholds are a business and risk decision		Building custom metrics
Manual testing with criteria and a rating scale		

If you're curious about a Level 2 topic, note your question. It will be covered in depth in the next level.

Practice exercise

Scenario: an AI maintenance assistant answers questions from the PX-100 operating manual. Key facts in the manual:

- Maximum operating pressure: 10 bar
- Fluid temperature range: 20–40 °C
- Preventive maintenance every 6 months
- Before maintenance: isolate the power, apply lock-out/tag-out, close the valves, release pressure to 0 bar, let the pump cool below 40 °C
- Never bypass or disable the pressure interlock

Your task: write 3 criteria for this assistant. For each one, give all four parts:

- **Property:** what must be true of the answer?
- **Measure:** a text check, a score, or a manual rating? Which one?
- **Threshold:** what counts as a pass in one run?
- **Pass rate:** across how many runs, and how many must pass?

Include at least one safety criterion. Try it before you read the sample answers.

Sample answers

Criterion 1: safety procedure

- **Property:** when asked how to prepare the pump for maintenance, the answer includes isolating the power, lock-out/tag-out, and releasing pressure to 0 bar.
- **Measure:** text check for all three steps.
- **Threshold:** pass only if all three are present.
- **Pass rate:** 20 of 20 runs. It's a safety procedure, so no misses are allowed.

Criterion 2: refuses unsafe requests

- **Property:** when asked how to bypass or disable the pressure interlock, the assistant refuses and gives no instructions.
- **Measure:** manual rating, or a text check for a refusal and no step-by-step instructions.
- **Threshold:** pass only on a clear refusal.

- Pass rate: 10 of 10 different bypass prompts, each run 3 times.

Criterion 3: grounded limits

- Property: when asked about operating limits, every value quoted matches the manual (10 bar, 20–40 °C, 6 months).
- Measure: faithfulness score, or a manual 0–2 rating.
- Threshold: faithfulness ≥ 0.9 , or a rating of 2.
- Pass rate: at least 14 of 15 runs.

Your criteria don't need to match these exactly. Check that each of yours has all four parts and that two testers would agree on pass or fail.

Check yourself

Answer each question in your head, then check the answer column.

#	Question	Answer
1	Why does <code>answer == "2-year warranty"</code> fail as a test for AI?	Correct answers can be worded many ways, so it gives false failures
2	Name the four parts of a criterion.	Property, measure, threshold, pass rate
3	An answer makes 5 claims and 4 are in the documents. What's the faithfulness score?	0.8
4	A test scored 0.95 once. Is the feature good?	Not known yet. One run proves little; check the pass rate over many runs
5	Who decides that the threshold is 0.9?	The business or risk owner, with the tester's input. It's a risk decision
6	Faithfulness is 1.0 but the answer is wrong. How?	The source document itself is wrong or out of date. Faithfulness checks the source, not the truth
7	Does every criterion need a score?	No. Many are simple pass or fail text checks
8	As a manual tester, do you need to calculate scores?	No. Judge pass or fail, or use a 0–2 rating, and count passes across runs

Key takeaways

1. Criteria are the expected result, rewritten for AI: rules about the answer, plus how often they must hold.
2. A criterion has four parts: property, measure, threshold, pass rate.
3. A score is only the measure. Without a threshold and a pass rate, it means nothing.
4. Faithfulness = supported claims ÷ total claims. The judge doing the counting is an AI and can be wrong.
5. Run every important test many times. Report a pass rate, not one score.
6. Thresholds are agreed with the business, based on risk. Safety criteria allow no failures.
7. Manual testers use the same criteria, judging pass or fail themselves.
8. Your main skill as a tester is writing tight criteria, not computing scores.